Oscar Cortes-Azuero, PhD

Postdoctoral researcher. My research interests include the dynamics, evolution, and burden of climate-sensitive infectious diseases with pandemic potential, with the aim of optimizing prevention and intervention strategies through data-informed mathematical models.

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I. RESEARCH POSITIONS

Apr 2026 - Present **Postdoctoral Research Associate**, University of Cambridge, UK
Department of Genetics, advised by Henrik Salje

Oct 2021 - Apr 2026 **Research Assistant**, University of Cambridge, UK
Department of Genetics, advised by Henrik Salje

Sep 2019 - Jan 2020 **Research Intern**, Institut Pasteur, Paris, France
Advised by Henrik Salje in the Mathematical Modelling of Infectious Diseases Unit, headed by Simon Cauchemez

II. EDUCATION

2021-2026 **Ph.D, Infectious Disease Epidemiology**
The spread of Nipah and chikungunya viruses: implications for control.
University of Cambridge - Cambridge, UK.

2017-2021 **Engineering Degree**
Master's Degree Equivalent: applied mathematics, optimization, signal processing.
Ecole Centrale de Lille - Lille, France.

2015-2017 **Undegraduate track in Mathematics and Physics**
Selective two-year program designed for national competitive entrance exams to French *Grandes Écoles*: advanced calculus, linear algebra, theoretical physics.
Lycée Louis-le-Grand, Paris, France.

III. RESEARCH GRANTS AND AWARDS

2026 Temasek Foundation. One-year salary award ($75,000).

2025 Coalition for Epidemic Preparedness Innovations. Co-Investigator; PI: Henrik Salje ($66,000).

2024 Coalition for Epidemic Preparedness Innovations. Co-Investigator; PI: Henrik Salje ($40,000).

2024 University of Cambridge. Travel award to attend the Global Young Scientists Summit in Singapore 2025 (£1,600).

2023 Jesus College, University of Cambridge. Jesus College Postgraduate Research Fund (£550).

IV. POLICY AND PUBLIC HEALTH IMPACT

2024 - Present **Consultant for Valneva Austria GmbH.**
Feasibility assessment of phase IV clinical trials for the IXCHIQ chikungunya vaccine in Brazil.
Design and revision of protocols for a randomized controlled trial and a test-negative case-control trial.
Preliminary evaluation of ongoing trials.

2025-2026 **Collaboration with the Coalition for Epidemic Preparedness Innovations**
Modeling the impact of climate and demographic changes on future demand for chikungunya vaccines to inform the *CEPI 3.0* strategy.

2024-2025 **Collaboration with the Coalition for Epidemic Preparedness Innovations**
Modeling the potential impact of medical countermeasures on Nipah virus outbreaks to inform logistical planning and future policy for pharmaceutical developers and local Ministries of Health.

V. PUBLICATIONS

Peer-reviewed publications

6. **Cortes-Azuero, O.** *et al.* Modeling optimal deployment strategies for Nipah vaccines and monoclonal antibodies. *Vaccine*. **84,** 128688 (2026).
5. **Cortes-Azuero, O.** *et al.* The epidemiology of chikungunya virus in Brazil and the potential impact of vaccines: a mathematical modelling study. *Lancet Infect. Dis.* **26**, 406-416 (2026).
4. Manuli, E. R., ..., **Cortes-Azuero, O.** *et al.* Transmission dynamics of Oropouche virus in Latin America and the Caribbean. *Nat. Med.* 1–10 (2026).
3. Cracknell Daniels B., ..., **Cortes-Azuero, O.** *et al.* A framework for risk–benefit analysis of vaccines approved through accelerated pathways. *Nat. Health* **1**, 120–128 (2026).
2. **Cortes-Azuero, O.** *et al.* The Genetic Diversity of Nipah Virus Across Spatial Scales. *J. Infect. Dis.* jiae221 (2024).
1. Kostandova, N., ..., **Cortes-Azuero, O.** *et al.* A systematic review of using population-level human mobility data to understand SARS-CoV-2 transmission. *Nat. Commun.* **15**, 10504 (2024).

Preprints in review

2. Abanda, N. N., ..., **Cortes-Azuero, O.** *et al.* The national burden from O'nyong'nyong and chikungunya viruses in Cameroon. *medRxiv.* 2026.06.24.26356335 (2026).

1. **Cortes-Azuero, O.** *et al.* The impact of climate and demographic changes on future chikungunya burden and the potential role of vaccines: a mathematical modelling study. *medRxiv.* 2026.02.16.26346397 (2026).

Comments & Perspective pieces

1. Salje, H. & **Cortes-Azuero, O.** The deadly potential of chikungunya virus. *Lancet Infect. Dis.* **24**, 442–444 (2024).

VI. PRIZES AND HONORS

2026 Jointly awarded a *Highly Commended Cambridge Award for Research Impact and Engagement* for work on chikungunya burden, immunization strategies, and risks and benefits of vaccines. University of Cambridge - Cambridge, UK.

2024 *Jesuan Award* for the support provided to the community in welfare and student supervision. Jesus College, University of Cambridge - Cambridge, UK.

VII. ORAL AND POSTER PRESENTATIONS

Invited talks

2025 Pandemic and Epidemic Threat Analysis Lab, School of Public Health, University of California, Berkeley - Berkeley, USA.

2025 Grubaugh Lab, School of Public Health, University of Yale - New Haven, USA.

2025 Infectious Disease Dynamics Group, Bloomberg School of Public Health, Johns Hopkins University - Baltimore, USA.

2025 Center for Epidemic Research and Modeling Seminar, Saw Swee Hock School of Public Health, National University of Singapore - Singapore.

2025 School of Public Health, The University of Hong Kong - Hong Kong, China.

2024 Kraemer Lab, Pandemic Sciences Institute, University of Oxford - Oxford, UK.

2024 Center for Mathematical Modelling of Infectious Diseases Seminar, London School of Hygiene & Tropical Medicine - London, UK.

2024 Panel Discussion: Wellbeing in a Warming World, Cambridge Climate Society - Cambridge, UK.

2022 BatOneHealth Seminar - Online.

Contributed talks and posters

2025 American Society for Tropical Medicine & Hygiene Annual Meeting - Toronto, Canada: two talks and a poster.

2025 Dengue Endgame Annual Summit - Syracuse, US: poster.

2025 Global Young Scientists Summit - Singapore: poster.

2024 American Society for Tropical Medicine & Hygiene Annual Meeting - New Orleans, USA: poster.

2023 EPIDEMICS, 9th International Conference on Infectious Disease Dynamics - Bologna, Italy: poster.

2023 EMBL-EBI & Cambridge Symposium - Hinxton, UK: talk.

2023 Infectious Disease Dynamics Conference - Ambleside, UK: poster.

2022 American Society for Tropical Medicine & Hygiene Annual Meeting - Seattle, USA: talk.

2021 EPIDEMICS, 8th International Conference on Infectious Disease Dynamics - Online: talk.

2021 18th Ecology and Evolution of Infectious Diseases meeting - Online: poster.